DUK Climate Scenario Valuation

A concise guide for a two-minute investment-team demonstration

01 The investment question

Question. How does Duke Energy's long-term enterprise value change when Current Policies climate effects are included, relative to a climate-agnostic reference case?

Target	Reference	Horizon	Output
Current Policies	Climate-agnostic baseline	2050	Conditional enterprise-value sensitivity

1	2	3	4	5
Define the question Target scenario, reference scenario, horizon	**Transmit risk** Transition, physical, macro / supply	**Build annual FCFF** Reference FCFF less the three annual pressures	**Discount and compare** Target EV versus reference EV	**Use with controls** Attribution, scenario range, data and assumption flags

What the headline means

Relative EV impact	$$(EV_{target} - EV_{reference}) / EV_{reference}$$

A scenario differential. It is not a probability, expected return, or share-price target.

For DUK under the selected comparison, the model reports **-4.74%**. The useful questions are whether the direction survives alternative scenarios, which channel drives the difference, and how sensitive the result is to the explicitly disclosed assumptions.

Client interpretation	Read this as a transparent long-term valuation sensitivity. Do not read it as 'DUK will fall 4.74%.'

End-to-end calculation architecture

The model is a sequence of transformations. Each stage has a defined input class, unit, calculation and control.

02	From scenario selection to decision output

Step	Operation	Input / calculation	Unit	Control
1	Select counterfactual	Target scenario, reference scenario, horizon, issuer and Low/Base/High case	Scenario definition	No calculation if target and reference are not explicit
2	Load observed issuer data	SEC FCFF components and PP&E; EPA Scope 1/2/3; facility-to-FEMA EAL matches	USD, tCO2e, EAL rate	Source status and as-of date retained
3	Build annual scenario paths	NGFS carbon-price path; scenario GDP and supply shock; CIE chronic pathway	USD/tCO2, %, annual index	Linear interpolation between governed observations
4	Calculate absolute target and reference costs	Run transition, acute, chronic and macro formulas for both scenarios	USD per year	Target and reference retained separately
5	Form incremental pressure	Pressure_k(t) = Cost_target,k(t) - Cost_reference,k(t)	USD per year	One driver belongs to one pillar only
6	Create annual FCFF paths	Reference baseline less transition, physical and macro/supply pressure	USD per year	Pre-floor and post-floor FCFF both retained
7	Discount target and reference	Forecast-period PV plus discounted terminal value	Enterprise value, USD	Same WACC and growth framework for both paths
8	Attribute and diagnose	Shapley contribution, EV floor, denominator sensitivity, proxy use	% and flags	RANKABLE / CAUTION / EXCLUDE gate
9	Interpret for investment	Scenario comparison, dominant driver, peer comparison and research gap	Research conclusion	No automatic trade or share-price target

Driver registry and double-counting rule

Pillar	Included	Explicitly excluded
Transition	Scope 1/2/3 emissions-price incidence after one pass-through adjustment	Physical damage, GDP and supply-chain shock
Physical	Acute asset damage plus chronic labour-productivity conversion	Carbon price and macro demand
Macro / supply	GDP and supply-chain shock scaled to issuer dependency	Carbon-price cost and physical damage

Transition-risk calculation

The engine computes target and reference emissions-price cost separately for every forecast year.

03	Emissions, carbon price, incidence and pass-through

Emissions path (1)

$$E_i(t) = E_i(2024) \times (1 - d)^{(t-2024)}$$

E_i is reported Scope i emissions; d is the demonstration annual decline assumption.

Gross Scope 1 (2)

$$G_{1,s}(t) = E_1(t) \times P_s(t) \times \text{coverage}$$

$P_s(t)$ is the annual NGFS scenario carbon price. Coverage represents the portion facing the modeled policy incidence.

Gross Scope 2 (3)

$$G_{2,s}(t) = E_2(t) \times P_s(t) \times \text{coverage} \times \text{scope2 incidence}$$

Scope 2 incidence is separated from reported emissions because contractual electricity treatment is not observed.

Gross Scope 3 (4)

$$G_{3,s}(t) = E_3(t) \times P_s(t) \times \text{scope3 incidence}$$

Scope 3 is a value-chain sensitivity, not an assertion that the issuer pays a carbon tax on all reported Scope 3 emissions.

Net cost (5)

$$C_s(t) = [G_{1,s}(t) + G_{2,s}(t) + G_{3,s}(t)] \times (1 - \text{pass-through})$$

Pass-through is applied exactly once inside each scenario's net cost.

Incremental pressure (6)

$$\text{Transition}(t) = C_{\text{target}}(t) - C_{\text{reference}}(t)$$

The model does not use target emissions times target price alone; it subtracts the full reference-scenario net cost.

DUK base demonstration parameters

Input	Value / source	Classification	Role
Scope 1/2/3	79.8m / 0.254m / 27.6m tCO2e; EPA / issuer evidence	Source-backed	Starting emissions
Carbon-price path	NGFS annual target and reference paths	Scenario	Price differential
Policy coverage	75%	Demo assumption	Scope 1/2 policy incidence
Pass-through	50%	Demo assumption	Customer cost recovery
2050 transition pressure	$30.3m	Derived	Target net cost minus reference net cost

Physical-risk calculation

Acute asset damage and chronic productivity pressure are calculated separately before aggregation.

04	Facility evidence, scenario scaling and financial conversion

Facility EAL (7)	$EAL_{DUK} = \sum[w_j \times \text{FEMA county building } EAL_j]$
	EPA facilities are geocoded to FEMA counties. The weights create an issuer facility evidence rate; DUK has 48 matched facilities out of 49.
Acute loss (8)	$\text{Acute}_s(t) = \text{Net PP\&E}; \times \text{covered share} \times EAL_{DUK} \times (1-\text{recovery}) \times \text{acute factor}_s \times \text{ramp}(t)$
	The FEMA EAL rate is observed context. Covered share, recovery and scenario acute factor are demonstration conversions.
Chronic loss (9)	$\text{Chronic}_s(t) = \text{Reference FCFF}(t) \times \text{labour share} \times \text{FCFF conversion} \times \text{CIE loss}_s(t)$
	CIE provides the scenario labour-productivity pathway. Labour share and the FCFF conversion rate are demonstration assumptions.
Physical pressure (10)	$\text{Physical}(t) = [\text{Acute}_{target}(t)+\text{Chronic}_{target}(t)] - [\text{Acute}_{reference}(t)+\text{Chronic}_{reference}(t)]$
	The reference physical cost is subtracted. Baseline is not silently assumed to equal target risk.

What is observed and what is assumed

Layer	Observed / scenario evidence	Demonstration conversion	Output
Acute	49 DUK facilities; 48 FEMA matches; emissions-weighted county building EAL	Covered PP&E; 65%; recovery 50%; scenario factor; annual ramp	Annual acute FCFF pressure
Chronic	CIE annual productivity-loss pathway; reference FCFF	Labour cost share and productivity-to-FCFF conversion	Annual chronic FCFF pressure
DUK 2050	High facility-match confidence: 98%	Base demonstration case	$79.0m physical pressure

Interpretation	The $79.0m result is not directly observed physical damage. It is a modeled FCFF pressure created by combining facility evidence and scenario pathways with disclosed financial-conversion assumptions.

Macro, annual FCFF and valuation path

The broader scenario shock is kept outside transition and physical risk, then all three pillars enter one annual cash-flow bridge.

05	Macro transmission, annual projection and DCF controls
Issuer dependency (11)	Dependency = bound[0.50 + SEC CapEx/CFO, 0.50, 2.00]
	This makes the issuer transformation depend partly on observed operating structure; the mapping itself remains a demonstration rule.
Macro loss (12)	Macro_s(t) = Reference FCFF(t) x [GDP loss_s + supply shock_s] x elasticity x dependency x ramp(t)
	GDP and supply-chain shocks come from the selected scenario; elasticity and the transformation rule are assumptions.
Incremental macro (13)	Macro(t) = Macro_target(t) - Macro_reference(t)
	Carbon-price cost and physical damage are excluded from this pillar by the driver registry.
Reference path (14)	Reference FCFF(t) = FCFF_2025 x (1 + g)^(t-2025)
	The prototype uses terminal growth g for the annual reference path; this is explicitly labelled as a demonstration assumption, not issuer guidance.
Target path (15)	Gross target FCFF(t) = Reference FCFF(t) - Transition(t) - Physical(t) - Macro(t)
	The engine retains the gross value before applying the zero cash-flow floor.
Floor control (16)	Target FCFF(t) = max[0, Gross target FCFF(t)]
	Any clipped amount is returned as unapplied cost. A floor or denominator-sensitive case receives an EXCLUDE ranking flag.
Discounting (17)	PV_s = sum_t FCFF_s(t)/(1+WACC)^t + [FCFF_s(T)(1+g)/(WACC-g)]/(1+WACC)^T
	DUK uses a displayed WACC of 5.12% and terminal growth of 2.00% in this governed demonstration release.

Annual audit trail and diagnostics

The headline can be reconstructed from annual cash flows; boundary and data-quality conditions remain visible.

05B	Reconciliation fields, valuation diagnostics and ranking gate

Annual audit fields returned by the engine

Field	Purpose	Review question
Reference baseline FCFF	Shows the unstressed annual path	Is the growth assumption appropriate?
Three annual pressures	Separates transition, physical and macro/supply	Does attribution reconcile and avoid overlap?
Gross and post-floor FCFF	Exposes nonlinear clipping	Is the result a boundary stress?
Discount factor and annual PV	Makes timing visible	Are WACC and timing internally consistent?

Valuation diagnostics

Diagnostic	Definition	Decision consequence
Symmetric EV difference	2 x (EV_target - EV_reference) / (|EV_target| + |EV_reference|)	Reduces one-sided denominator distortion
Unapplied cost	Negative gross FCFF removed by the zero floor	Any positive amount triggers EXCLUDE
Denominator sensitivity	Target/reference EV ratio outside 0.25x to 4.0x	Treat percentage as a boundary result
Physical proxy	Issuer facilities unavailable and cross-issuer FEMA median used	Ordinary issuer ranking is excluded
Attribution residual	Total EV difference less three Shapley contributions	Must reconcile to numerical tolerance

Ranking gate

Status	Trigger	Permitted use
RANKABLE	No floor, denominator, proxy or material coverage flag	Same-scenario comparison, subject to model limits
CAUTION	Facility match coverage below 75%	Analyst review before comparison
EXCLUDE	FCFF floor, unapplied cost, denominator sensitivity or proxy	Do not include in ordinary issuer ranking

Equation map and input classification

A compact cross-reference for the calculations demonstrated on the platform.

06	Three transmission channels and the valuation equation

Transition	Net transition pressure(t) = Net cost_target(t) - Net cost_reference(t)
	Gross cost is built from emissions x scenario carbon price, energy cost and transition investment. Customer pass-through is applied once inside each scenario's net cost.
Physical	Physical pressure(t) = Acute(t) + Chronic(t)
	Acute pressure links facility-level FEMA evidence to covered PP&E; and an assumed damage/recovery conversion. Chronic pressure uses the scenario's long-run heat and productivity pathway.
Macro / supply	Macro pressure(t) = Reference FCFF(t) x scenario shock(t) x issuer dependency
	The scenario shock captures GDP and sector/supply transmission. Issuer differentiation uses observed operating structure, including the SEC CapEx-to-cash-flow relationship.
Annual FCFF	FCFF_target(t) = FCFF_reference(t) - Transition(t) - Physical(t) - Macro(t)
	The engine returns every annual baseline, pressure, pre-floor and post-floor FCFF, discount factor and present value.
Enterprise value	EV_s = sum[FCFF_s(t) / (1 + WACC)^t] + TV_s / (1 + WACC)^T
	Target and reference paths use the same valuation framework. The result stops at enterprise value until net debt and other equity-bridge inputs are governed.

Data versus model assumptions

Class	Examples in the DUK demonstration	How to describe it
Observed / source-backed	SEC CFO, CapEx, interest, tax and PP&E; EPA emissions; 48 of 49 facilities matched to FEMA EAL	Company or facility evidence
Scenario pathways	NGFS/CIE carbon, GDP, energy and physical pathways	Conditional scenario inputs
Prototype assumptions	Coverage, pass-through, recovery, covered PP&E;, damage conversion, macro elasticity and terminal growth	Demonstration assumptions - not issuer-observed
Derived	Annual channel pressures, target/reference FCFF, discounted enterprise values and attribution	Model outputs

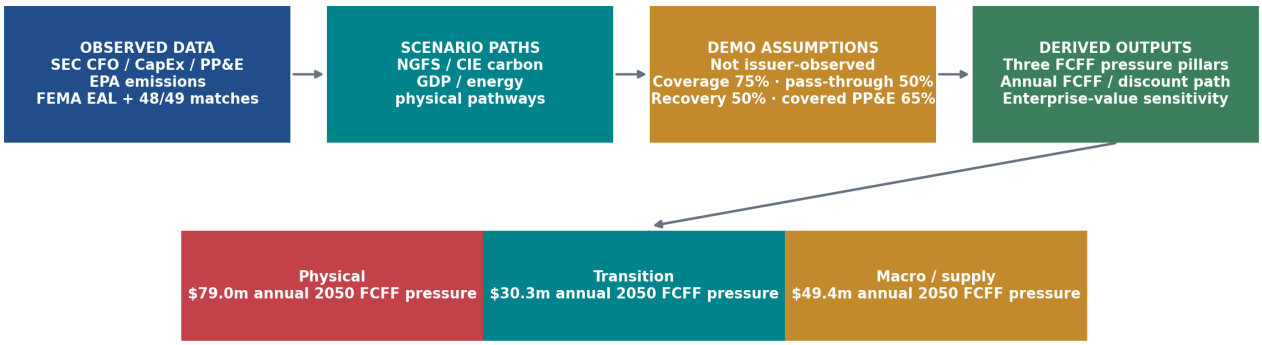
DUK worked example

Current Policies versus climate-agnostic baseline | Base demonstration assumptions | 2050

07 From source evidence to FCFF pressure and enterprise value

DUK model data provenance

Each output is traceable to observed data, scenario pathways, explicit demo assumptions, or derived calculations.



DUK demo readout: −$158.6m annual 2050 FCFF change and -4.74% enterprise-value sensitivity.

Enterprise value only; equity value and share-price impact are not calculable without a governed EV-to-equity bridge.

Prototype demonstration | Source-backed inputs + explicitly labelled assumptions (not issuer-observed) | EV sensitivity, not probability or price target | CFP-2026-003-R17-20260804T234500Z

Reference FCFF 2050	Transition	Physical	Macro / supply	Target FCFF 2050
$2.513bn	-$30.3m	-$79.0m	-$49.4m	$2.354bn

The annual 2050 bridge is $2.513bn - $30.3m - $79.0m - $49.4m = $2.354bn. Physical pressure is the largest of the three terminal-year channels. The model then discounts the full annual target and reference paths, not only the terminal-year change.

Reference EV	Target EV	EV difference	Relative sensitivity
$50.13bn	$47.76bn	-$2.37bn	-4.74%

Important boundary	This is an enterprise-value sensitivity. Equity value and share-price impact are not calculable until a governed EV-to-equity bridge is available.

Investment use and interpretation

The prototype is most useful as a comparative research framework, not as a standalone Climate VaR estimate.

08	Decision use and the two-minute demonstration sequence

Decision question	Model contribution	Required analyst judgement
Scenario resilience	Compare Current Policies, Net Zero, Below 2C and Delayed Transition using the same engine	Which pathway is material to the investment thesis?
Driver attribution	Separate transition, physical and macro/supply contributions	Is the modeled driver economically credible for the issuer?
Peer comparison	Apply one scenario and one assumption case consistently across comparable issuers	Are business models, regulation and data quality sufficiently comparable?
Research priorities	Expose facility, emissions, pass-through and conversion gaps	Which missing evidence should be obtained from management or vendors?

Two-minute viewing sequence

0:00-0:25	Define DUK, the target/reference comparison and the 2050 horizon.
0:25-0:45	Read -4.74% as a conditional enterprise-value sensitivity, not a return forecast.
0:45-1:25	Show the three FCFF channels and identify physical pressure as the largest DUK contributor.
1:25-1:52	Show the annual FCFF-to-EV bridge and explain the shared WACC/terminal-value framework.
1:52-2:15	Separate source-backed inputs from assumptions and close with investment use.

Recommended close	Use the model to compare scenarios, identify the dominant value driver and test assumptions. Do not use it as an automatic trade, audited valuation or replacement for the analyst's forecast.